

TRILOKESH RANJAN SARKAR

Data Analyst | Machine Learning | Data Visualization | Big Data

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Summary

Data Analyst with 1 year of experience in Big Data, SQL, Python, machine learning, ANPR, and OCR. Skilled in developing OCR pipelines for digitizing handwritten documents and leveraging NLP for data extraction, reducing manual effort by 80%. Proficient in creating actionable insights using **Power BI, Looker Studio, and SQL**. Strong background in **data cleaning, visualization**, and supporting business decisions. Holds a Post Graduate Program in Data Science and is passionate about turning complex data into clear, impactful solutions.

Experience

Web Spiders

Kolkata, West Bengal

Data Analyst

10/2024 - Present

- Worked with **U.S. based client's data team** to **train and improve an AI model** that analyzes **transaction patterns**, leading to an **8.3% boost in r2 Score** and improvements in **forecasting**.
- Used **advanced SQL & regex queries** to perform **data cleaning, matching, and extraction** on **credit card transaction data**, helping identify **false negatives** and remove **false positives** for more accurate results.
- Used **Power BI & SQL** to visualize trends and present actionable insights that supported strategic financial planning.
- Utilised **pre-trained models** for **sentiment analysis** of in-store customer **facial expressions**, helping track emotions and boost engagement.
- Built **heat maps** from **video analytics** to spot **high-traffic areas** and **peak hours**, improving **store layout** and **staff planning**.
- Created a custom **PyTesseract** handwriting **OCR pipeline to digitize handwritten medical documents**. Additionally, developed OCR workflows that reduce **human intervention in the data extraction process by 80%** by using **natural language processing (NLP)** for named entity recognition (NER) and regex matching.
- Developed **ANPR** system using **YOLO v8** for fast and accurate license plate detection.
- Built dashboards to monitor email campaign performance (open rate, reply), enabling data-driven decisions that led to a **15% increase in engagement** and **12% boost in conversions**.
- **Tech Stack:** SQL, Python, Power BI, AWS Cloud Server, Regex, NLP.

Praxis Business School

Kolkata, West Bengal

Research Associate – Generative Artificial Intelligence

12/2023 - 05/2024

- Tested **adversarial robustness** of a Generative AI by comparing **FGSM** and **Carlini-Wagner attacks** to evaluate model security against adversarial threats.
- Used the **Carlini-Wagner** and **FGSM** attacks to evaluate the security of the model. Improved the robustness and efficiency of the model through optimized LLM fine-tuning under resource constraints.

Education

Praxis Business School

Kolkata, West Bengal

Post-Graduation Program in Data Science | CQPI: **6.2** / 8

2023 - 2024

- **Ranked in the top 5%** of a highly competitive academic program, demonstrating strong dedication, consistency, and excellence.

B. P. Poddar Institute of Management and Technology

Kolkata, West Bengal

Bachelors of Computer Application | CGPA: **8.8** / 10

2020 - 2023

Certifications

[Databricks Fundamentals Accreditation](#) — [Databricks](#)

[SQL Certificate](#) — [Hacker Rank](#)

[Streaming Data with AWS Kinesis and Lambda](#). — [DataCamp](#)

[Generative AI Fundamentals Accreditation](#) — [Databricks](#)

PUBLICATION

arXiv.org

Comparing FGSM, Carlini-Wagner Attacks, and the Function of Distillation as a Defense Mechanism to Assess Adversarial Robustness

2023 - 2024

<https://arxiv.org/abs/2404.04245>

- **Published in Arxiv Tech's Research Gate** after peer review by top GenAI researchers, recognizing the work's contribution to adversarial defense techniques in machine learning.
- Provided a thorough **evaluation framework** comparing model vulnerabilities and defense effectiveness, offering actionable insights for secure model deployment..